

Helion: the next diagnostic decision

Which procedure should the engineer run next?

Lower investigation cost and turnaround, subject to the same diagnostic completeness and safety requirements.

1. Does manufacturing context improve **fault prediction**?
2. Do predictions improve **diagnostic selection** beyond rules?
3. Can dispatch improve the existing **scheduling examples**?

The engineer owns diagnosis and closure. A proposed shift coordinator owns dispatch. Coexisting faults remain separate diagnostic obligations.

Source: Client brief; expanded user scope; research_v1 benchmark.

The experiment and its comparators

Offline pipeline: validate, train, score, replay, schedule and report.

Policy	Next-procedure choice
Mock incumbent	Triggered work first, then remaining completeness work
CT-first rules	Consider triggered and completeness work together
Four heuristics	Coverage/cost with prevalence or one of three models

Shared standard: supported findings for all seven mechanisms, no unresolved contradictions and satisfied audit obligations.

One attempt per procedure. Unfinished work stays pending. A low model probability cannot establish absence. Unresolved intact-sample work blocks destructive SEM.

Source: Pipeline README; MOCK-ENG-001; benchmark §2. Same costs, report scope and paired report draws.

Data and operating assumptions

Supplied synthetic data

916 rejected stacks, seven fault labels

653 / 126 / 137

training / validation / test

Test: **6** die-crack cases and

2 cases with coexisting faults

Assumed operations

8 technicians and 8 diagnostic engineers

Skills, one-day calendars and reservations

Dollar rates, task phases and procedure error rates

Resource dollars per attempt: CT \$120 acoustic \$200 electrical \$600 IR \$150 SEM \$1,800

Some allowed acceptance readings directly encode generated faults. The already-inspected test cohort provides retrospective evidence.

Still needed: real procedure histories, qualified SOP, complete audit findings and observed costs.

Source: Validation record; benchmark evidence boundary; SYN-OPS-001. Staff and costs are invented assumptions.

Fault prediction on the test cohort

137 retrospective test cases. Seven binary logistic heads allow coexisting faults.

Input comparison	Macro AP $\uparrow$	Log loss $\downarrow$
Constant training prevalence	0.145	0.402
Inspection/acceptance	0.504	0.231
Full allowed context	0.497	0.251
Manufacturing/package only	0.189	0.408

0.497 is macro average precision: a ranking summary averaged equally across seven faults. It does not mean 49.7% of diagnoses are correct.

Full-minus-inspection AP: -0.007 ; paired 95% interval $[-0.058, +0.047]$. The small AP difference is uncertain.

Source: predictive_metrics.csv and predictive_paired_comparisons.csv, test rows. Synthetic shortcuts limit generalisation.

ML matches CT-first diagnostic outcomes

Base test replay: 137 cases $\times$ 100 paired report replications per policy.

Policy	Consumed USD/case	Evidence complete	Correctly complete
Mock incumbent	\$1,995.55	53.53%	48.41%
CT-first rules	\$1,994.63	53.54%	48.42%
Full-model heuristic	\$1,994.63	53.54%	48.42%

Full model = CT-first on cost, completion, errors and nominal duration in all **13,700 paired runs**.

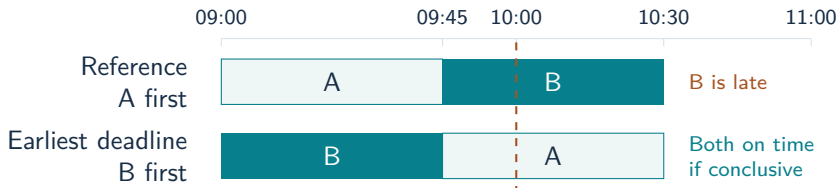
Full minus mock: **−\$0.92/case**; 95% lot-bootstrap interval [**−\$2.84, \$0.00**]. CT-first captures the same result.

Every case counts, including unfinished investigations. Full model: \$747.59/case in known pending procedures. Unknown review continuation remains unpriced.

Source: decision_metrics.csv; decision_uncertainty.csv; verification.json. Synthetic resource costs, not fab savings.

Dispatch changes which milestone is met

Case D: resources unavailable before 09:00. B's warpage milestone is 10:00; A's underfill milestone is 11:00.



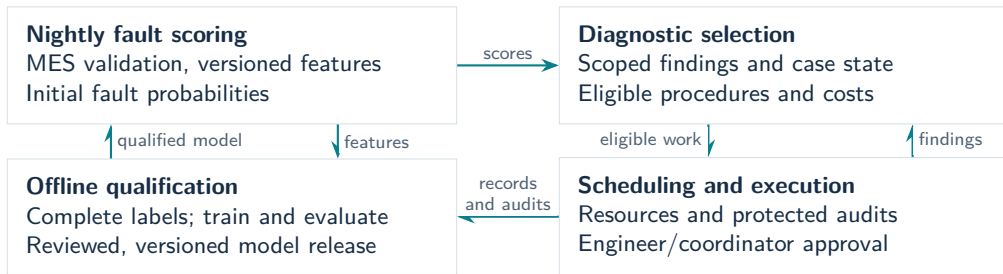
Both orders cost \$240 and finish at 10:30. Staff phases, equipment and specimen exclusivity remain feasible.

Inconclusive findings stay unresolved. Scheduling covers the existing 24-hour fixtures only; cohort hours are nominal procedure totals.

Source: scheduling.json; SYN-OPS-001 Case D. Scoped milestones depend on conclusive findings. Actual dispatch is unknown.

Proposed fab operation and feedback

Offline prototype implemented. Diagram: proposed fab integration and qualification.



Monitor: batch health, input drift, audited errors, queue age, inconclusives and overrides.

Fallback: rules/manual review for failed imports, unqualified contexts or stale calendars. Incomplete audits remain incomplete.

Source: Design blueprint and architecture; pipeline README. Yield Engineering owns outcomes, lab operations calendars, IT ingestion.

What the evidence supports next

No incremental ML diagnostic value demonstrated over CT-first rules under these assumptions.

Prediction Investigate weak-fault errors, available features and regularisation using training-lot validation.

Decision Qualify the real SOP, useful alternative paths, examination scope and continuation costs.

Evidence Replace operational assumptions and collect representative complete audits plus fresh prospective evidence.

Retaining rules remains valid. Any shadow study or controlled pilot needs qualification. No policy is selected for operational release.

Source: Benchmark limitations and next evidence; client brief audit requirement. Improvement experiments are proposed.